



Linktree

Projecting Crop Yields based on ESM Emulation

Gina Roberg Charles Wang Qilong Zou Nilay Menon
groberg@ucsd.edu czwang@ucsd.edu q1zou@ucsd.edu nimenon@ucsd.edu

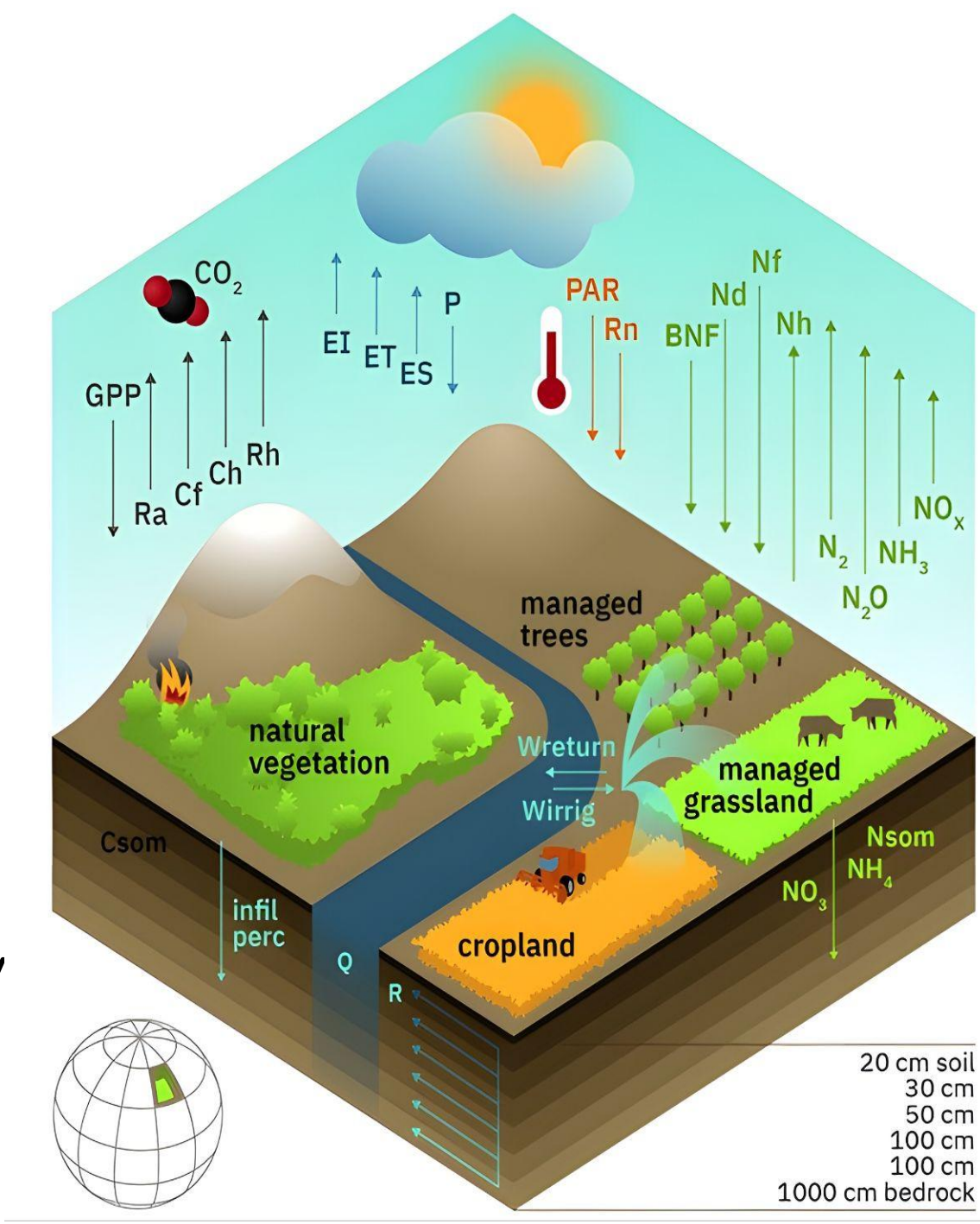
UC San Diego™
HALICIOĞLU DATA SCIENCE INSTITUTE

Mentor: Duncan Watson Parris
dwatsonparris@ucsd.edu

Introduction

Climate change is one of the most pressing global challenges, affecting ecosystems, economies, and human livelihoods. Understanding its impact is crucial for mitigating risks and developing sustainable adaptation strategies. However, assessing these impacts remains a challenge due to the complexity of climate systems and the computational cost of traditional Earth system models and Impact model.

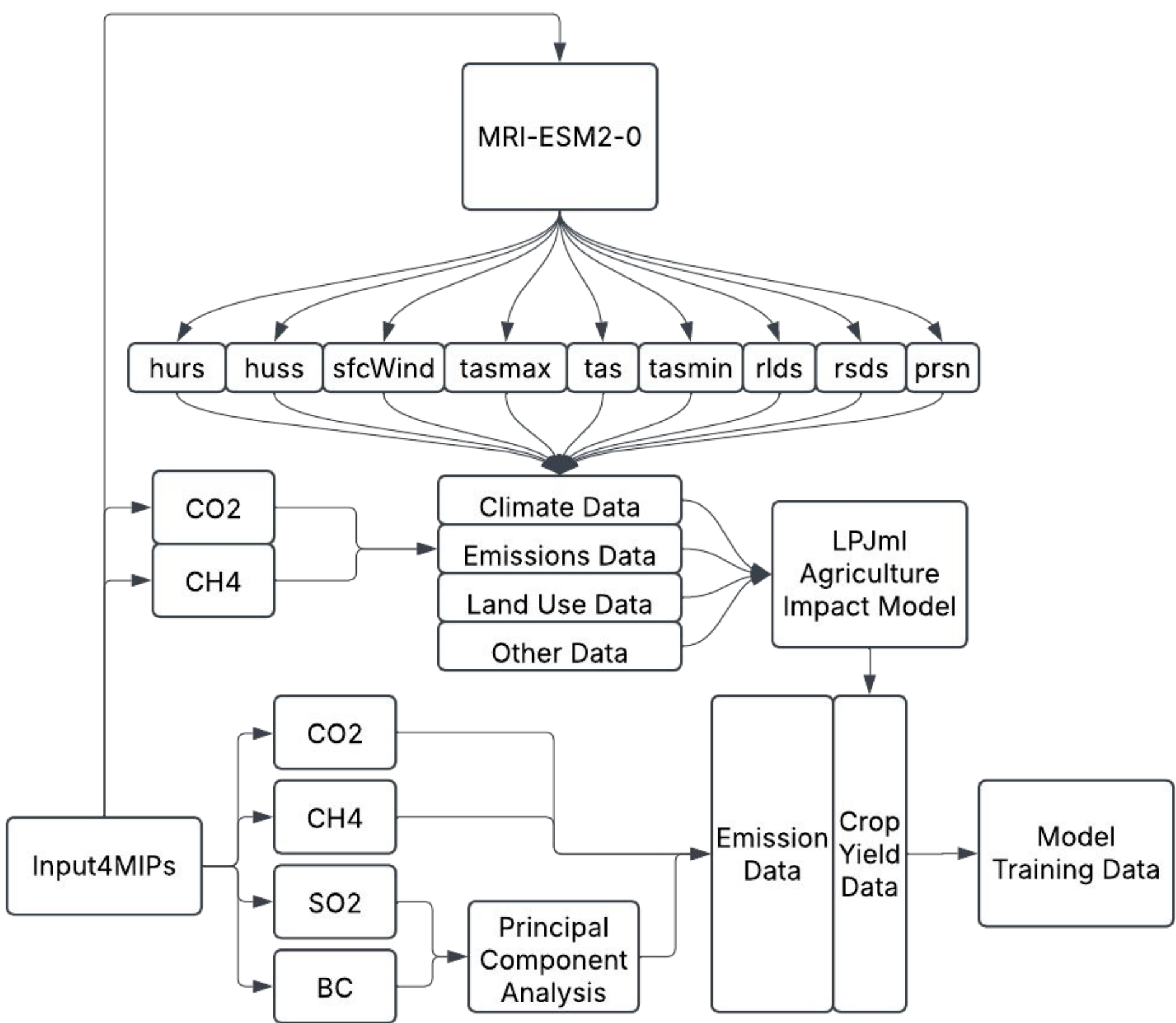
- Earth System Model emulators can predict climate variables based on various greenhouse gas emissions efficiently
- Resource-Intensive:** Earth system models require high-performance computing and can take weeks to run a single simulation.
- Difficult Interpretation:** Results of Earth System Model do not clearly convey how human life is impacted by these climate variables



Our project leverages machine learning to rapidly and accurately estimate climate impacts on agriculture, making scenario exploration more efficient and accessible than ever before.

Data

- Coupled Model Intercomparison Project Phase 6 (CMIP6):** Collaborative framework for climate modeling that provides standardized climate simulations. *MRI-ESM2-0* is one of the ESM developed by Meteorological Research Institute.
- Inter-Sectoral Impact Model Intercomparison Project (ISIMP):** Similar to CMIP6, but aims to provide standardized climate impact simulations across different sectors. LPJmI is one of the impact model built on top of *MRI-ESM2-0* for the agriculture sector.



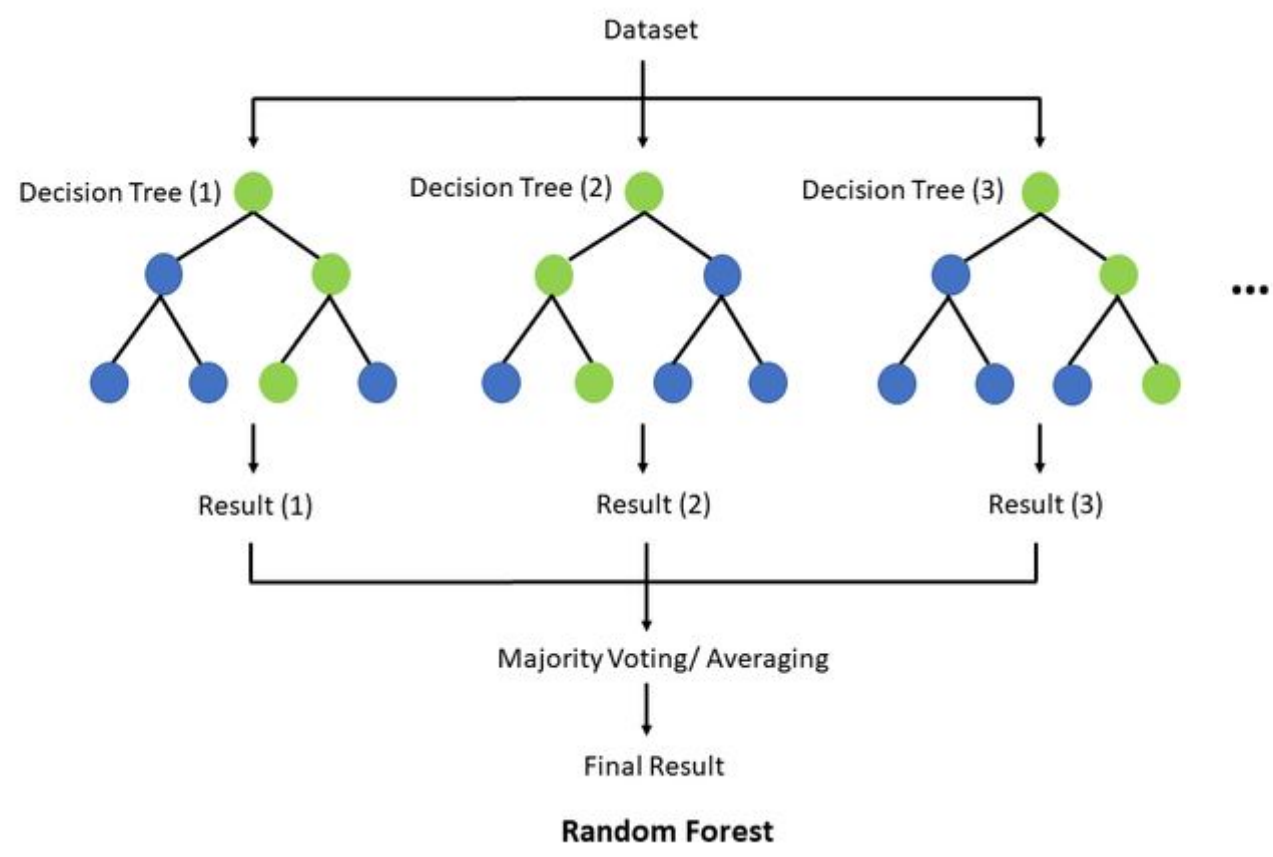
Methods - Gaussian Process Regression

Gaussian Process Regression (GPR) is a non-parametric Bayesian method that models relationships in data without assuming a fixed equation. Instead, it uses a flexible, kernel-based approach to make predictions while naturally handling uncertainty.

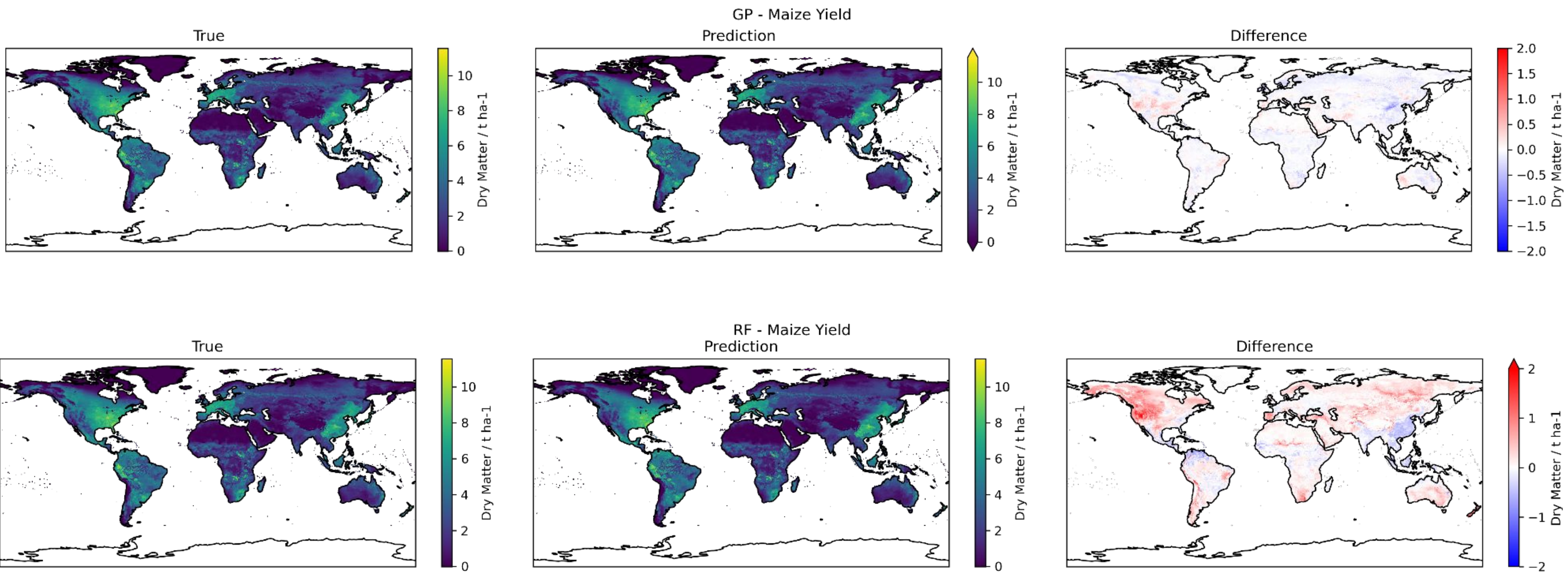
- Uncertainty Quantification:** Provides confidence intervals to assess prediction reliability
- Nonlinear Modeling:** Able to capture complex relationships between emissions and crop yields
- Data Efficiency:** Performs well with limited emissions data compared to Deep Learning Models
- Automatic Relevance Determination (ARD):** Allows automatic identification of the most influential emission factors during training

Methods - Random Forest

- Implemented using the Earth System Emulator (ESM) library,
- Leverages an ensemble of decision trees to model complex climate-yield interactions
- Output data averaged over latitude and longitude to handle missing values and reduce noise in data



Predictions

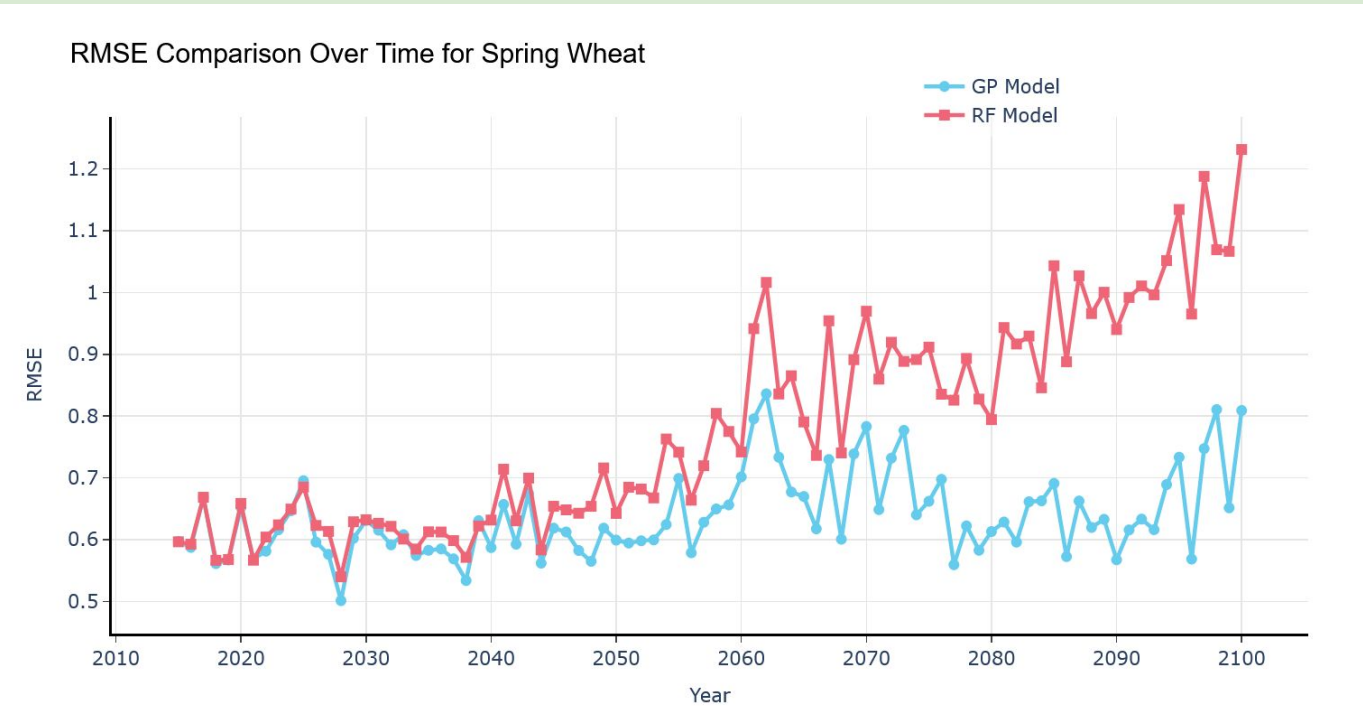
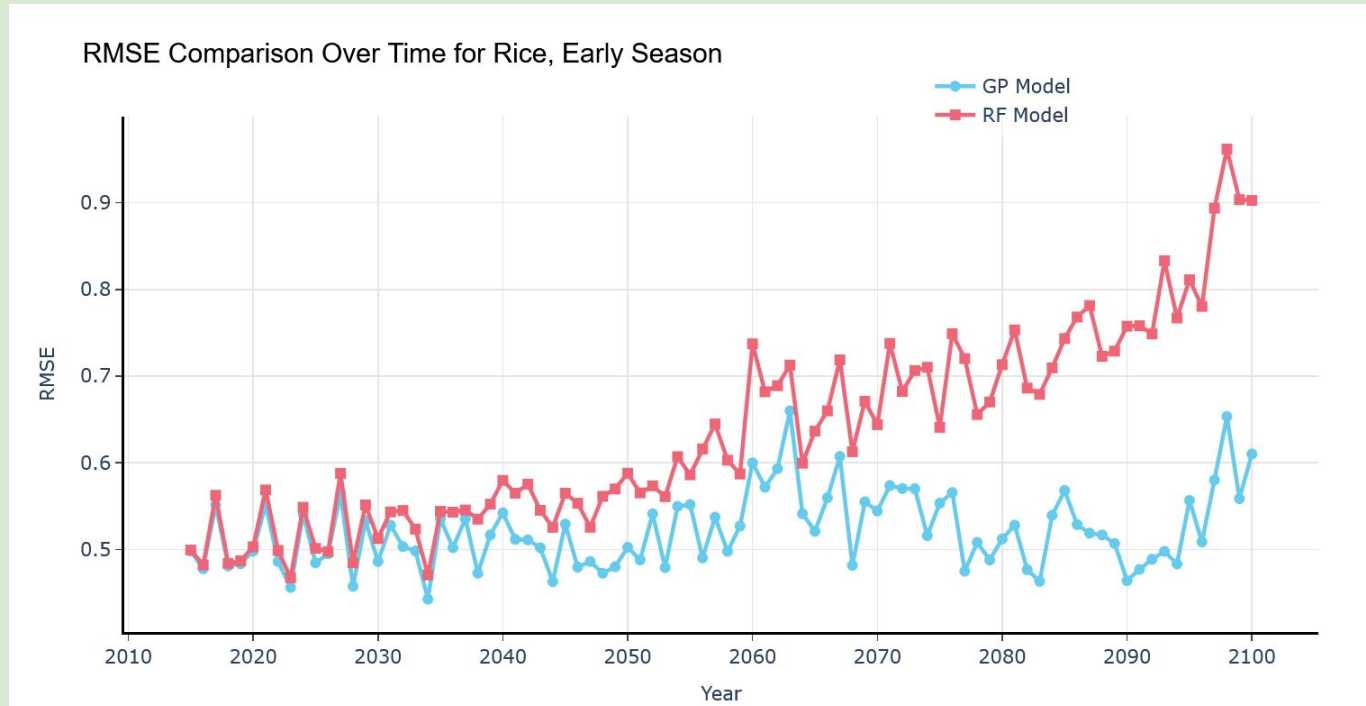


The maps above show the mean maize yield predictions for the years 2050-2100. The first column represents the ground truth (LPJmI crop yield), while the second column displays predictions made using our models. The third column shows the difference between the predicted and true values.

Result

To compare our emulator predictions, we calculate the root-mean square error (RMSE), where it includes a weighting function that accounts for the decreasing grid-cell area toward the poles.

Crops	RMSE at 2050		RMSE at 2100		RMSE 2045-2055		RMSE 2090-2100		RMSE 2050-2100		RMSE average last 20y	
	GP	RF	GP	RF	GP	RF	GP	RF	GP	RF	GP	RF
Maize	0.5649 6	0.6044 5	0.6151 3	0.7651 2	0.5526 3	0.5722 6	0.5375 8	0.6650 1	0.5521 7	0.6266 1	0.1607 7	0.4060 0
Rice - 1st growing season	0.5142 9	0.5879 5	0.6223 8	0.9027 6	0.5135 6	0.5686 9	0.5475 8	0.8289 9	0.5459 9	0.7053 5	0.2042 6	0.6038 5
Rice - 2nd growing season	0.3886 7	0.4641 3	0.4971 5	0.7004 7	0.3926 7	0.4330 0	0.4384 0	0.6685 1	0.4240 0	0.5479 8	0.1357 3	0.4790 2
Soybean	0.8344 6	0.9243 7	1.0664 7	1.3934 3	0.8716 5	0.9153 6	0.9444 6	1.2186 9	0.9395 4	1.0921 3	0.2887 1	0.7765 3
Spring wheat	0.5859 9	0.6424 5	0.8105 5	1.2315 9	0.6073 3	0.6814 6	0.6844 2	1.0587 6	0.6673 9	0.8918 0	0.2730 2	0.8046 2
Winter wheat	0.6526 6	0.6961 6	0.9100 3	1.2370 7	0.6751 6	0.7267 1	0.7918 7	1.0473 6	0.7475 3	0.8900 9	0.2846 7	0.6977 9



- GP model has less predictive error but is sensitive to missing data and computationally more demanding
- RF model is computationally more efficient but has more predictive error
- Models can guide crop selection, irrigation planning, and policy decisions.
- Soybeans are riskier in extreme climates, while maize and rice are more predictable.

Discussion / Future Direction

- Findings can support early warning systems and adaptation policies,** helping governments and farmers adjust agricultural practices to mitigate climate-related yield losses.
- Adapt state-of-the-art machine learning models** from current research to enhance performance.
- Experiment with diverse data sources**—such as soil data or local weather—to improve prediction accuracy, particularly for challenging crops like soybeans.
- Extend the methodology beyond agriculture** to other climate-sensitive sectors for broader impact.